Continuous-time Kogut–Susskind expansion: corrected overlap bounds and an open explicit uniform threshold

Correction (2026-10-02): The former volume-uniform physical-gap claim $\Delta_{\mathrm{KS}} \geq \frac{4}{3}g^2\hbar c/a$ for $g^2 \geq 388$ is **withdrawn as a conclusion of this note**. The adjacent count roots the first plaquette at the test link and misses histories reaching that link later. The proposed general-tree count does not bound the required marked overlap integrals, and incorrectly excludes repeated plaquettes. The shift measure must include every excitation interval; its endpoint term costs up to $4n$, not n. The claimed true-vacuum correlation bound was also not established. Sections 2–4 correct these steps and label the remaining numerical estimates. The earlier identification of T2's constant with $4(1-\theta)$, its volume-uniform limit 4, and relative error $O(g^{-4})$ is withdrawn here. The existing existential strong-coupling theorem is unaffected.

Later-note scope pointer: mass-gap-openings §1 replaces the all-coupling requirement by the weak scaling-region requirement; its §2, Opening 1 curvature comparison concerns Wilson Euclidean measures and does not repair this Hamiltonian expansion.

The results established here are the electric-history norm bound, the corrected overlap estimate conditional on a factorizing history gas, and a finite-volume physical-gap bound. For $SU(3)$ on a periodic cubic spatial lattice with $N_s \geq 4$, writing $P = |\mathcal{P}| = 3N_s^3$,

$$\Delta_{a,L}^{\mathrm{phys}} \;\geq\; \left(\frac{8}{3}g^2 - \frac{12P}{g^2}\right)\frac{\hbar c}{a}, \qquad L = N_s a.$$

In particular $g^4 \geq 9P$ gives $\Delta_{a,L}^{\mathrm{phys}} \geq \frac{4}{3}g^2\hbar c/a$. This bound is extensive and supplies **no volume-uniform threshold**. At each fixed volume the physical electric-loop gap is recovered as $g \to \infty$. The continuous-time construction below is a candidate route to uniformity, not a completed gap theorem. The older numerical thresholds are retained with their corrected status in §4; no continuum limit is taken.

1. The expansion and the finite-volume result

Units $\hbar c/a = 1$. Work on the gauge-invariant subspace of $L^2(SU(3)^{\mathcal{E}})$. Write $H = H_E - W + C$ with

$$H_E = \frac{g^2}{2}\sum_{\ell}(-\Delta_\ell), \qquad W = \frac{1}{g^2}\sum_{p} w_p, \qquad w_p = \operatorname{tr} U_p + \operatorname{tr} U_p^\dagger, \qquad C = \frac{6P}{g^2}.$$

Thus the magnetic potential is $(2/g^2)\sum_p(3 - \operatorname{Re}\operatorname{tr} U_p)$. In a Peter–Weyl basis, including matrix indices and invariant vertex intertwiners, the electric eigenvalue is

$$E(\sigma) = \frac{g^2}{2}\sum_{\ell} C_2(r_\ell) \geq \varepsilon|S(\sigma)|, \qquad \varepsilon = \frac{2g^2}{3},$$

where $S(\sigma)$ consists of nontrivial link representations and $4/3$ is the least nonzero $SU(3)$ Casimir. Also $\|w_p\| \leq 6$.

Correction (2026-10-02), topology and sector: Gauss's law excludes a vertex incident to exactly one excited link. Nonabelian invariant networks can branch, so describing every support as a union of flux loops was too strong. A nonempty finite graph with minimum degree at least two contains a cycle. For the cubic lattice with $N_s \geq 4$, its shortest cycle has four edges; hence $|S| \geq 4$. Short periodic lattices require a separate treatment: winding cycles at $N_s = 2, 3$ can be shorter. The four-link bound concerns the **physical sector**; the electric gap on the unrestricted Hilbert space is ε.

The constant function Ω_0 is the unique electric vacuum. The plaquette character $\operatorname{tr} U_p$ is gauge invariant, nonzero, orthogonal to Ω_0, and an eigenvector with energy 4ε. Therefore the

physical electric gap is exactly 4ε. Since $\|W\| \le 6P/g^2$, the min–max principle bounds the shifts of the lowest two eigenvalues of $H_E - W$ by $6P/g^2$ each. Consequently

$$4\varepsilon - \frac{12P}{g^2} \;\le\; \Delta_{a,L}^{\mathrm{phys}} \frac{a}{\hbar c} \;\le\; 4\varepsilon + \frac{12P}{g^2}.$$

The scalar C cancels from the gap. This proves the lead's bound and $\Delta_{a,L}^{\mathrm{phys}}/(g^2\hbar c/a) \to 8/3$ at fixed N_s. A nonpositive lower bound is simply uninformative. Finite-volume compactness and positivity improvement supply the unique physical true ground state as proved in T1 §2 (repository proof read); this assertion alone is not uniform in volume.

After removing C, Duhamel's formula is

$$e^{-t(H_E-W)} = \sum_{n\ge 0} \int_{0<\tau_1<\cdots<\tau_n<t} e^{-(t-\tau_n)H_E} W e^{-(\tau_n-\tau_{n-1})H_E} \cdots W e^{-\tau_1 H_E}\, d\tau_1 \cdots d\tau_n.$$

Correction (2026-10-02): These operator summands and the resolved history amplitudes are not asserted to be positive. The interaction w_p takes both signs; the bounds use absolute values.

Insert $1 = \sum_S P_S$ between factors, where P_S projects onto the exact excited link set. These projections factor over links and commute with gauge transformations. For a vacuum-to-vacuum history let $S_0 = S_n = \emptyset$, $x_k = \tau_{k+1} - \tau_k$ for $1 \le k < n$, and

$$A(\gamma) = \sum_{k=1}^{n-1} |S_k| x_k.$$

The history amplitude satisfies

$$|w(\gamma)| \le \left(\frac{6}{g^2}\right)^n e^{-\varepsilon A(\gamma)}.$$

Outside the plaquette links S_k cannot change, and $\mathrm{links}(p_k) \subseteq S_{k-1} \cup S_k$: multiplication on a trivial link produces a fundamental or antifundamental excitation. There are at most 16 possible next excited sets per insertion. An irreducible history with no intervening vacuum has $|S_k| \ge 4$.

The intended polymer support consists of the excited link intervals, with insertion vertices recording which plaquette couples their ends. Independent histories on disjoint link factors have factorizing amplitudes; summing their chronological interleavings gives the product of their internal ordered integrals. A complete polymer construction must specify components and endpoint compatibility consistently with these insertion vertices. The hard-core representation

$$Z(t) = \langle \Omega_0, e^{-t(H_E-W)}\Omega_0\rangle = \sum_{\{\gamma_i\}\ mcompatible} \prod_i w(\gamma_i)$$

is the proposed organization. The finite-volume bound above does not require it. For §§2–3 assume such an organization with the history majorant, $|S_k| \ge 4$ inside each polymer, and changes only on the four links of each insertion; the missing general count remains necessary even under these assumptions.

2. The corrected convergence criterion

For an established hard-core polymer gas on a measure space, a sufficient Kotecký–Preiss condition is

$$\int_{\gamma' \not\sim \gamma} |w(\gamma')| e^{a(\gamma')}\, d\gamma' \le a(\gamma).$$

The measure-space statement follows from Poghosyan–Ueltschi, §2, Assumptions 1–2 and Theorem 2.1 (passage read): hard-core interactions have stability function zero. For finite volume and time one also needs the integrability assumption of that theorem. Uniform local bounds,

when proved, control the corresponding rooted cluster sums. This abstract criterion does not itself identify a Hamiltonian spectral gap.

Set

$$a(\gamma) = n(\gamma) + \lambda A(\gamma), \qquad \delta = \varepsilon - \lambda > 0, \qquad h = \frac{6e}{g^2}.$$

The tilted majorant at fixed shape and internal times is $b(\gamma) = h^n \exp(-\delta \sum_k |S_k| x_k)$. Let $r_\ell(\gamma)$ count maximal excitation intervals of link ℓ, and $T_\ell(\gamma)$ be their total length. Then

$$\sum_\ell T_\ell = A(\gamma), \qquad \sum_\ell r_\ell \le 4n(\gamma),$$

because each interval begins at an insertion and an insertion touches four links.

Correction (2026-10-02), shift measure: For one interval of length x'_k on link ℓ in γ', the allowed translations overlapping γ have measure at most

$$T_\ell(\gamma) + r_\ell(\gamma) x'_k.$$

Indeed, an interval of length L_i in γ gives a translation interval of length $L_i + x'_k$; take the union bound over all r_ℓ intervals. The former $T_\ell + x'_k$ expression omitted this multiplicity.

Define the majorant over **all** connected shapes reaching a test link, without requiring their first plaquette to contain it:

$$Q_\lambda = \sup_{\ell_0} \sum_{\text{shapes } \exists k : \ell_0 \in S_k} (n-1) h^n \prod_{k=1}^{n-1} \frac{1}{\delta |S_k|}.$$

It bounds the same integral summed over marked intervals k containing ℓ_0. Its length-marked version is at most $Q_\lambda/(4\delta)$, since, term by term,

$$\int_0^\infty e^{-\delta |S| x}\, dx = \frac{1}{\delta |S|}, \qquad \int_0^\infty x e^{-\delta |S| x}\, dx = \frac{1}{(\delta |S|)^2} \le \frac{1}{4\delta} \frac{1}{\delta |S|}.$$

The chronological internal-time domain has been enlarged to the full positive orthant. Summing the overlap union bound over links gives

$$\int_{\gamma' \not\sim \gamma} |w(\gamma')| e^{a(\gamma')}\, d\gamma' \le Q_\lambda A(\gamma) + \frac{Q_\lambda}{4\delta} \sum_\ell r_\ell(\gamma) \le Q_\lambda A(\gamma) + \frac{Q_\lambda}{\delta} n(\gamma).$$

Thus the corrected sufficient conditions are

$$Q_\lambda \le \lambda, \qquad Q_\lambda \le \delta.$$

The former second condition $Q_\lambda \le 4\delta$ lost a factor four. These are conditional sufficient inequalities, not a proof that the actual Q_λ is finite or small.

3. What the shape counts actually establish

Adjacent growth with the first plaquette anchored. Fix a link ℓ_0 and restrict to histories whose first plaquette contains it and whose later insertions touch the current excited set. There are four initial plaquettes; S_1 is forced. Each later insertion has at most $4|S_{k-1}|$ plaquette choices and 16 support choices. The preceding interval integral cancels $|S_{k-1}|$, giving

$$u = \frac{64h}{\delta} = \frac{384e}{g^2 \delta}.$$

For this restricted class the weighted sum with multiplier $n-1$, denoted B_λ rather than Q_λ, obeys

$$B_\lambda \le 4h \sum_{m \ge 1} m u^m = \frac{24e}{g^2} \frac{u}{(1-u)^2}, \qquad u < 1.$$

This is a valid overcount of the restricted class.

Correction (2026-10-02), missing roots: Even within adjacent growth, a polymer can start away from ℓ_0 and reach it later. Thus B_λ does not bound Q_λ. Rooting at the first insertion that touches ℓ_0 requires controlling both the earlier and later history. No such estimate is supplied here. Even a gas restricted to adjacent growth still needs this additional root estimate for KP.

General shapes and the proposed tree count. A later insertion can start a disjoint piece that is subsequently bridged. Join consecutive insertions touching each link during a maximal excitation interval. For a connected insertion graph and any spanning tree T, distinct edges assigned to a link cover nonoverlapping subintervals, so

$$\sum_{e\in T} |\tau_{i(e)} - \tau_{j(e)}| \le A(\gamma), \qquad e^{-\delta A(\gamma)} \le \prod_{e\in T} e^{-\delta|\tau_{i(e)}-\tau_{j(e)}|}.$$

This energy inequality is valid with a specified link assignment. Each insertion has at most eight predecessor/successor link slots. It does not establish the originally quoted complete tree majorant.

Correction (2026-10-02), general count withdrawn: The old expression

$$u_{\mathrm{gen}} = 32 \cdot 3 \cdot 16\,\frac{h}{\delta} = \frac{9216e}{g^2\delta} = 24u$$

was presented as rigorous. A child insertion may use the **same** plaquette as its parent: creating and then annihilating a single plaquette loop is already an example. A fixed parent link therefore allows four plaquettes, not three. More substantially, the rooted link/interval multiplicities and time-length marks in §2 were not counted. A tree edge need not be a consecutive *global* time gap, so replacing every chronological denominator by a tree-edge integral $1/\delta$ does not justify the length-marked bound. The assertion that the overlap criterion follows with $Q \le \delta$ under this tree count is withdrawn. Changing three to four alone would give $32u$ in this formal expression, but is not a repair of the missing proof. A usable general estimate must bound both the interval-marked integral and its length-marked counterpart, or give another proved overlap majorant. Neither $24u$ nor $32u$ is a certified general counting factor.

4. Numerical estimates and the spectral implication

Write $\lambda = \varepsilon(1-\theta)$, $\delta = \varepsilon\theta$, $0 < \theta < 1$. The exact restricted factor is

$$u = \frac{576e}{\theta g^4}.$$

For $u \le 1/2$, the proved restricted bound is $B_\lambda \le 96e\,u/g^2$. Requiring this bound to be at most both λ and δ is implied by

$$g^4 \ge \frac{1152e}{\theta}, \qquad g^8 \ge \frac{82944e^2}{\theta\min(\theta, 1-\theta)}.$$

The old $3132/\theta$ is a rounded upper approximation to $1152e/\theta$, and 6.13×10^5 approximates $82944e^2$. The second condition is not universally weaker as $\theta \to 1$. These inequalities control B_λ only. They do not certify KP or a gap.

θ	former adjacent threshold g^2	sufficient integer g^2 for the displayed restricted inequalities	candidate decay rate 4λ (unproved for H)
1/2	79	80	$\frac{4}{3}g^2$
1/4	112	112	$2g^2$
1/10	177	177	$2.4g^2$

Correction (2026-10-02), constants and table status: $79^2 = 6241$ is below $2304e$, so 79 was rounded down even for the restricted sufficient condition. The old formal general factor is exactly

$$24u = \frac{13824e}{\theta g^4}, \qquad 13824e \simeq 37578.17,$$

not 37584. Its half-factor condition alone is $g^4 \geq 27648e/\theta$. The following records the old table and the correct rounding of that **formal condition alone**, not thresholds for the actual general gas or Hamiltonian:

θ	former "rigorous" threshold g^2 (withdrawn)	integer satisfying $24u \leq 1/2$ only
1/2	388	388
1/4	548	549
1/10	867	867

The old accompanying $g^8 \geq 1.47\times10^7/[\theta(1-\theta)]$ was likewise a formal algebraic estimate from the unproved count; it is not retained as a convergence condition. Neither table is a physical-gap table. Rates in its last column would be in units $\hbar c/a$ if independently established.

True-vacuum spectral lemma (finite volume). Let ψ_0 be the normalized true ground state, E_0 its energy, and A a bounded gauge-invariant multiplication observable. Put $f_A = (A - \langle\psi_0, A\psi_0\rangle)\psi_0$. Suppose for every such observable (including ones supported on the whole finite lattice)

$$0 \leq \langle f_A, e^{-t(H-E_0)} f_A\rangle \leq C_A e^{-mt}, \qquad t \geq 0.$$

Then $\Delta_{a,L}^{\text{phys}} \geq m\hbar c/a$ in the present dimensionless time convention.

Proof. The spectral measure μ_A of f_A is positive. If it had positive mass on $[0, m-\eta]$ for some $\eta > 0$, its Laplace transform would be at least $\mu_A([0, m-\eta])e^{-(m-\eta)t}$, contradicting the assumed bound. Thus every f_A has zero spectral weight below m. These vectors are dense in the physical ground-state complement: $\psi_0 > 0$ almost everywhere, and any physical f can be approximated by $A_M\psi_0$ with $A_M = (f/\psi_0)\mathbf{1}_{\{|f/\psi_0|\leq M\}}$, which is bounded and gauge invariant. Centering gives the asserted density. Boundedness of spectral projections extends their vanishing to that complement. This proves the lemma. In infinite volume the same implication needs the specified vacuum GNS representation and a dense observable domain.

Correction (2026-10-02), missing correlation estimate: The lemma's hypothesis has not been established by this expansion. Correlations with external Ω_0 are not automatically true-vacuum correlations. For each finite volume one can obtain the latter by placing observables in the interior of a long time slab, dividing by $Z(2T+t)$, and sending $T \to \infty$. Explicitly, for A centered in the true vacuum,

$$\frac{\langle\Omega_0, e^{-TH}A^*e^{-tH}Ae^{-TH}\Omega_0\rangle}{\langle\Omega_0, e^{-(2T+t)H}\Omega_0\rangle} \longrightarrow \langle\psi_0, A^*e^{-t(H-E_0)}A\psi_0\rangle.$$

This follows from the rank-one ground-state spectral projection and $\langle\psi_0, \Omega_0\rangle > 0$; it gives no quantitative decay rate. A polymer proof must control the normalized, connected insertion

expansion uniformly in the boundary distance T and volume. Also, the four-link energy bound for individual physical histories does not by itself prove that the union of every correlation cluster has area at least $4t$. The required spanning-cluster estimate and observable prefactors must be proved. Consequently $m = 4\lambda$ remains a candidate rate, and no volume-uniform $\gamma \to 4$ or ratio $18/\gamma = 4.5$ follows here. Only the fixed-volume limit in §1 is proved.

5. What remains of the time-discretized comparison

Correction (2026-10-02): "Yarotsky's theorem gives a threshold 10^{101}" and "this beats it by giving 10^2" were too strong. Yarotsky's theorem provides an existential smallness threshold, not that number; the latter is the repository's crude proof-extraction estimate in the former threshold calculation (now incorporated below). Its assumptions are reproduced below so the comparison is checkable, but the estimate is **heuristic**, not a certified sufficient threshold of this note.

Normalize by $\varepsilon = 2g^2/3$. Group the three plaquette potentials based at one site: their total norm is at most $54/g^4$. If one assumes a space-time support entropy c^n with $c = 80e$, a per-support-site activity bound ρ^n, and the sufficient condition $\sum_{n\geq 1}(ce\rho)^n \leq 1$, then $\rho \leq 1/(2ec) = 1/(160e^2)$. Assume further the site activity estimates

$$\rho_J = e^{-t_0}, \qquad \rho_I = 2e\, t_0 \beta\, e^{64 t_0}.$$

The 64 comes from the proposed shadow-volume estimate $|\Lambda_0|^2$ with $\Lambda_0 = \{0,1\}^3$, $|\Lambda_0| = 8$. The insertion prefactor follows by minimizing $2\alpha e^{t_0\beta/\alpha}$ over $\alpha > 0$, at $\alpha = t_0\beta$. Thus $t_0 \geq \log(160)+2$ controls ρ_J, and choosing $t_0 = 7.1$ would require

$$\beta \leq \frac{e^{-454.4}}{160e^2\, 2e \cdot 7.1}, \qquad \log\beta \lesssim -465, \qquad \log g_0^2 = \tfrac{1}{2}(\log 54 - \log\beta) \simeq 234.5.$$

This is roughly 10^{102}; the former 10^{101} quotation was only an order-of-magnitude description. The extraction still needs a proved conversion of the full configuration weights and their multiplicities into the assumed support-site activity bound; it is not a quantitative application of Yarotsky's theorem proved here. In particular e^{64t_0} is the cost in this assumed shadow estimate, not an unavoidable cost of every time-discretized method.

The established result used instead is the local-perturbation theorem as applied in strong-coupling-uniform-gap §§1–2 (repository hypothesis check read; Yarotsky's Theorem 1 at the passage level recorded there): for sufficiently small $54/g^4$ there is a positive volume-uniform gap, with unspecified constants. Retaining exact link decay in continuous time is a plausible way to improve constants, but the counts and boundary estimates missing above prevent any proved numerical improvement in this note.

The older direct-expansion comparison used an illustrative coordination factor 12 and $C \sim e$ to estimate $g^4 \gtrsim 54 \cdot 12e$, or g^2 of order 40–70. No applicable Kirkwood–Thomas or Datta–Kennedy bound with these constants has been proved here; the papers were inspected at metadata level. The Wilson Euclidean character expansion concerns a different transfer operator. As a one-loop orientation, $n \simeq (g_{\text{weak}}^{-2} - g_{\text{strong}}^{-2})/0.0966$ for $SU(3)$ gives about 10^3 doublings if $g_{\text{weak}}^{-2} = 100$, whether the hypothetical strong boundary is $g^2 = 70$ or 10^{101}, and about 21 for $g_{\text{weak}}^2 = 1/2$ and $g_{\text{strong}}^2 = 70$. These are extrapolations, not certified boundaries or RG trajectories.

6. Consequence for STATE

This note no longer supplies an explicit volume-uniform KS threshold $g^2 \geq 388$, the adjacent value 79, or a uniform gap coefficient approaching 8/3. Its proved outputs are the fixed-volume bound of §1, the conditional overlap criterion of §2, and the restricted count of §3. The numerical

tables are restricted or formal estimates, with their status stated at each use. T2 remains supported by the separate existential strong-coupling theorem. Any catalog or downstream use of this note's old uniform thresholds, including the target-box tolerances, needs to inherit this correction. An explicit continuous-time uniform threshold requires a complete marked general-history count and a normalized true-vacuum correlation bound. This correction changes no other note, no STATE file, and makes no weak-coupling or continuum claim.